INTERIM FEDERAL SPECIFICATION
FIRE TRUCK (PUMPER), GASOLINE OR DIESEL ENGINE (4x2, 6x4) 20,000 POUNDS G.V.W. AND OVER

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers gasoline and diesel engine driven, four-wheel, two rear wheel drive and six wheel, four rear wheel drive fire trucks. Vehicles procured under this specification are commercial type and are intended for use as a pumper.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commerically available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types. Fire trucks shall be as specified in the solicitation for offer (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

F-G-621 - Governors, Gasoline Engine; Centrifugal, Velocity and Air-vane, hydraulic and electronic.
VV-F-800 - Fuel Oil, Diesel.
VV-G-76 - Gasoline, Automotive.
ZZ-I-550 - Inner Tube, Pneumatic Tires.
ZZ-T-381 - Tires, Pneumatic, Vehicular (Highway).
O-E-915 - Extinguishers, Fire, Dry Chemical (Portable).
O-F-555 - Foam - Forming Liquids, Concentrated, Fire Extinguishing, Mechanical.
ZZ-H-451 - Hose, Fire, Woven Jacketed, Rubber or Cambric Lined, with couplings.
WW-C-621 - Coupling Assembly, Hose (Fire, Woven Jacketed, Rubber or Cambric - Lined and Unlined).
ZZ-H-421 - Hose and Hose Assembly, Chemical (Industrial and Fire Fighting).

Federal Standard:

Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a

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subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Service Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Laws and Regulations:

29 CFR 1926.601 - Motor Vehicles
40 CFR 85 - Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines
40 CFR 202 - Noise Emission Standards for Motor Carriers Engaged in Interstate Commerce
40 CFR 205 - EPA Transportation Equipment Noise Emission Control
49 CFR 393.11-96 - Parts and Accessories Necessary for Safe Operation
49 CFR 393.94 - Vehicle Interior Noise Levels
49 CFR 571.101-302 - Federal Vehicle Safety Standards

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

State of California, Vehicle Emissions Standards:

(Application for copies should be addressed to the California Air Resources Board, 1709 24th Street, Sacramento, California 95809.)

Tire and Rim Association Yearbook:

(Copies may be obtained from the Tire and Rim Association, Inc., 3200 West Market Street, Akron, Ohio 44313.)

Society of Automotive Engineers (SAE), Inc., Standards and Recommended Practices:

J245 - Engine Rating Code, Spark Ignition.
J350 - Spark Arrester Test Procedure for Medium Size Engines.
J551 - Measurement of Electromagnetic Radiation from Motor Vehicles (20-1000 MHz).
J552 - External Electromagnetic Radiation Suppressors.
J682 - Rear Wheel Splash and Stone Throw Protection-Commercial Vehicles.
J683 - Tire Chain Clearance-Trucks, Buses and Combinations of Vehicles.
J688 - Truck Ability Prediction Procedure.
J704 - Side Openings for Truck Power Take-Off.
J816 - Engine Test Code, Spark Ignition and Diesel.

(Copies may be obtained from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, Pennsylvania 15096.)

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National Fire Protection Association Standard:

No. 1901 -Automotive Fire Apparatus

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, Massachusetts 02210.)

American Welding Society

Standard Qualification Procedure

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, NY 10017.)

American Society of Mechanical Engineers

Boiler and Pressure Code, Welding Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Design. The fire engine furnished under this specification shall be basically the manufacturer's current design for the type specified. The allied equipment specified shall be complete with all necessary operating accessories customarily furnished together with such modifications and attachments as may be necessary to enable the vehicle to function reliably and efficiently in sustained operation. The design of the vehicle and optional equipment shall permit ready accessibility for servicing, replacement, and adjustment of component parts and accessories with minimum disturbance of other components and systems. After body equipment and accessories are added to the chassis for completion of the vehicle, the total weight of the finished product including body, equipment, accessories, driver, and payload shall not exceed the chassis manufacturer's gross axle weight rating (G.A.W.R.) or the gross vehicle weight rating (G.V.W.R.) for the chassis furnished. The apparatus shall be constructed with due consideration to the nature and distribution of the load to be sustained, and to the general character of the service to which the apparatus is to be subjected when placed in service. All design, operational and material features must fully comply with Federal Motor Vehicle Standards as stated in Public Law No. 89-563, including additional pertinent acts, amendments, and standards in effect at time of manufacture. The vehicle and apparatus shall also comply with the current, at time of manufacture, Standard No. 1901 of National Fire Protection Association, covering Automotive Fire Apparatus.

3.2 Materials. The materials shall be new with quality conforming to current engineering and manufacturing practices. Materials shall be free of defects, suitable for the intended service. Virgin materials are not required and recovered materials are not excluded in manufacturing.

3.3 Vehicle equipment and accessories. The vehicle components, assemblies, and accessories to be delivered under the contract shall meet or exceed the requirements of this specification. They shall comply with all Environmental Protection Agency Regulations, Federal Motor Vehicle Safety Standards and Federal Motor Carrier Safety Regulations applicable to the specified vehicle on the date of manufacture. All chassis items shall be as represented in the chassis manufacturer's current technical data, and special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's current technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, or on file by

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the company as being engineered for the particular application, prior to award of contract for a vehicle. The component parts of the vehicle need not be the products of the same manufacturer.

3.3.1 Special equipment. In addition to the standard vehicle and components specified herein, each vehicle shall be furnished when required by the solicitation for offer or contract, with special equipment as specified therein, when specified (see 6.1).

3.4 Vehicle lubrication. The vehicle, chassis components, and equipment shall be fully equipped with lubrication fittings in accordance with the manufacturer's standard practice.

3.5 Vehicle weight, rating, and dimensions.

3.5.1 Curb weight. Curb weight shall include weight of chassis and cab with all chassis attachments, accessories, equipment and the booster tank and body, including a full complement of fuel, lubricant and coolant.

3.5.1.1 Gross vehicle weight rating. Gross vehicle weight rating (G.V.W.R.) shall be specified (see 6.1) and shall include curb weight, operator's weight, plus the weight of the number of support personnel for which space is provided (weight shall be computed at 175 lbs. per person), payload and water.

3.5.2 Rating. Component and vehicular ratings, as published in the manufacturer's current technical data, shall not be arbitrarily raised to meet the requirements of this specification. Manufacturers shall provide a G.V.W.R. plate showing actual G.V.W.R. of the vehicle furnished.

3.5.3 Dimensions. Cab to axle (CA) dimension shall be specified when dimension is required for type and size of vehicle ordered. The gross vehicle weight of the truck shall be distributed with the following minimum on the front axle and shall not exceed the axle capacity.

Truck	Percent
Five man cab forward	35
Conventional cab - 4X2	20
Tilt cab - 4X2	25
Any type cab - 6X4	20

Manufacturer shall furnish vehicle with the shortest CA dimension available that is adequate for intended body and provides proper weight distribution. Maximum overall width shall not exceed 96 inches unless specifically waived.

3.5.3.1 Unless otherwise specified or required by stated CA dimension (see 6.1), wheel base shall be manufacturer's standard type of vehicle offered.

3.6 Frame. The frame shall be capable of withstanding the strain of rigorous service and shall be provided with adequate crossmembers so designed and constructed to support the gross weight of the vehicle with its load, personnel, power plant and all specified equipment. The frame, as supplied by the chassis manufacturer to the fire engine builder shall not be altered, either to be increased or decreased in length other than to remove excess length of frame, rails behind the rear spring hangars, which will not affect the original wheel base, without the permission of the contracting officer and then only after specific engineering details of the procedure have been furnished, along with a complete warranty for the life of the vehicle, of the work performed, barring accidents or collision.

3.6.1 Frame reinforcement. When fire trucks are to be equipped with aerial ladder, elevated platform, snorkel units, or when minimum resisting bending moments are specified (see 6.1), the frame design and material shall be ample to withstand

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the maximum stress imposed under extreme operating conditions. This may be accomplished by increased resistance to bending moment by reinforcing frame channels, use of deeper section to provide equivalent section modulus, or use of higher tensile strength steel frame rails. Frame shall provide sufficient structural strength to equal all loads imposed under all specified operational and environmental conditions.

3.7 Power unit, gasoline or diesel.

3.7.1 Power unit. Power unit furnished for the specified vehicle shall be the chassis manufacturer's standard, optional, or commercially available engine for the commercial model truck which meets or exceeds the required vehicle performance specified herein (see 3.8), at not more than engine manufacturer's recommended operating engine speed. It shall be of such design and construction that it will give an even flow of power at all r.p.m. without undue vibration, strain, or overheating of engine parts.

3.7.1.1 Engine starting. Unless otherwise specified in 3.28, engine, gasoline or diesel, shall start satisfactorily without devices that are not self-contained on the vehicle at temperatures down to -10° F. The vehicle manufacturer shall incorporate sufficient battery capacity to satisfy this requirement. Auxiliary heating or starting devices shall be provided at the option of the purchaser.

3.7.2 Gasoline engine. Unless otherwise specified (see 6.1), power unit shall be liquid cooled, internal combustion, four stroke cycle gasoline type, not less than six cylinders and shall conform to requirements as specified herein. Power unit shall provide efficient performance on low lead or unleaded regular grade gasoline (low lead gasoline contains 0.5 grams or less of lead per gallon). Vehicles for export shall be equipped with engine compression ratio most suitable for the indicated ultimate destination. Engines furnished shall be rated in accordance with SAE J245.

3.7.3 Diesel engine. When specified (see 6.1), the engine shall be a liquid cooled, compression ignition type, two or four stroke cycle with at least six cylinders. Engine shall develop the required horsepower and torque (see 3.8), and operate smoothly and efficiently when operated on diesel fuel conforming to the requirements of VV-F-800. Engines furnished shall be rated in accordance with SAE J816.

3.8 Performance.

3.8.1 Speed and gradeability. Speed, gradeability and acceleration requirements shall be met with the truck loaded to specified and rated G.V.W.R., unless drive train ratios are specified in the invitation for bid or contract. Gear ratios in transmission and axles shall be matched to provide a progressive shifting pattern throughout the complete range and shall provide required vehicle performance. Performance greater than the minimum may be specified to meet more rigid service demands and, if required, shall be specified under 3.8 in the invitation for bid or contract. All acceleration tests, with the exception of high speed gradeability test, shall be performed on level road.

3.8.2 Maximum road speed. The maximum road speed (true speed) at engine governed speed shall be no less than 55 m.p.h.

3.8.3 High speed gradeability. Vehicle shall be capable of ascending a 2% grade at 50 m.p.h. when calculated in accordance with SAE J688 (see 6.2).

3.8.4 Low speed.

3.8.5 Acceleration requirements. From a standing start through the gears, the vehicle shall attain a true speed of 35 m.p.h. within 25 seconds.

From a steady speed of 15 m.p.h., the vehicle shall accelerate to a true speed of 35 m.p.h. within 30 seconds. This shall be accomplished without moving gear selector.

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3.9 Governor. Engine shall be equipped with a suitable governor approved by the engine manufacturer, set to limit engine speed to maximum recommended operating speed and shall be sealed at the specified setting. Governor need not be stamped to show speed at which set. (Exception: When vehicle is delivered by driveaway, or when a break-in period is recommended by the manufacturer, the governor may be set at reduced r.p.m. to protect engine during delivery and/or break-in periods. In this case, the governor shall be stamped to indicate maximum r.p.m. to allow resetting to manufacturer's specification at end of break-in period or completion of driveaway delivery, as appropriate. (Exception: The manufacturer may reset governor, in which case governor need not be stamped.) Governors for gasoline engines must conform to the requirements of F-G-621.

3.10 Oil filter. The oil filter shall be of the full flow type, so constructed that no special equipment or skill will be required in changing the filtering element. Unless otherwise specified (see 6.1), the capacity of the gasoline engine oil filter shall be at least one quart, and minimum two quart capacity for diesel engines. The oil filter shall meet or exceed the engine manufacturer's lubrication requirements. When specified, an independently mounted by-pass type oil filter, with replaceable element, of the size recommended by the engine manufacturer shall be provided for use with diesel engines, in addition to the standard full flow filter.

3.11 Air filter. Unless otherwise specified (see 6.1), air filter shall be manufacturer's standard for engine offered. When equipped or specified with an oil bath type, oil capacity shall be not less than one quart.

3.12 Emission controls. Vehicles and engines shall comply with the regulations of the Environmental Protection Agency governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. California pollution control requirements shall be provided when specified (see 6.1), and on those vehicles destined for operation in the State of California. Export vehicles shall be furnished with manufacturer's standard pollution control requirements applicable to such vehicles.

3.13 Fuel system. The fuel system furnished shall conform with Federal Motor Carrier Safety Regulations 393.65 - 393.67 and include fuel tanks equipped with an electrical fuel level sensing device with gauge visible to the seated operator. Unless otherwise specified (see 6.1), the style tank(s) shall be manufacturer's standard for the vehicle offered. Fuel tank(s) of not less than 30 gallons total capacity is required. Truck filler opening shall be so located to provide ready access to filler opening when using standard type fuel delivery nozzle, without obstruction by any portion of mounted body or equipment.

3.13.1 Fuel system, gasoline engine. A carburetor or fuel injection device, mechanically or electrically-driven fuel pump and fuel filter(s) shall be furnished. When dual fuel tanks are furnished, a fuel tank selector valve identified to indicate which tank is furnishing fuel, shall be installed. A selective switch to operate fuel gauge readings shall be installed on the instrument panel adjacent to the fuel gauge. The switch positions shall be identified to indicate which tank reading is being taken.

3.13.2 Fuel system, diesel engine. A mechanically-driven fuel pump with all necessary regulatory and controlling attachments accessible and readily removable injectors and fuel filter(s) shall be furnished. When dual fuel tanks are furnished, a balance line shall be provided to equalize the amount of fuel in each tank.

3.14 Cooling system. The cooling system shall be liquid pressurized force circulation type. It shall consist of the necessary components of sufficient design and capacity to maintain the engine at optimum safe temperature under all operating conditions specified herein, without any loss of coolant. Optimum engine temperature shall be maintained with the vehicle loaded to the G.V.W.R. and continually operating at all driveable altitudes and grades in ambient temperatures ranging from -30° to

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125° F. The cooling medium shall be thermostatically controlled and shall be suitable for operating with permanent type of antifreeze solutions.

3.14.1 Drain outlets. Easily accessible drain outlets shall be provided on all vehicles to allow complete cooling system drainage.

3.14.2 Shutters. When specified (see 6.1), or when required to accomplish control of coolant and operating temperature, automatic thermostatic controlled shutters shall be provided.

3.15 Exhaust system. Each truck shall be equipped with an efficient muffler and exhaust system. The tailpipe shall be of such size to avoid undue increase in back pressure and shall be located in such a manner that entrance of exhaust gases into cab or body shall be minimized under all conditions of operation. Location of the tailpipe shall avoid undue heating of fuel system components, battery containers, and tires. When necessary, or required, adequate insulation or shielding shall be utilized. Discharge end shall be so directed to avoid undue disturbance of road dust and clogging. Tailpipe location may be specified (see 6.1). The exhaust system shall be manufacturer's heaviest-duty corrosion resistant type, securely fastened, and routed to protect system from hazards and shall be in conformance with Federal Motor Carrier Safety Regulation 393.83. When specified (see 6.1), a spark arrester type exhaust system having 80 percent arresting efficiency shall be furnished in accordance with SAE J350.

3.16 Power train.

3.16.1 Clutch. The manufacturer's heaviest duty clutch available for the engine and chassis model offered shall be furnished, with torque capacity exceeding maximum gross engine torque.

3.16.2 Transmission. Manufacturer's selective manual or automatic type transmission shall be provided. Input torque capacity of the transmission shall be at least equal to the maximum gross torque delivered by the engine. Transmission shall be provided with a standard SAE 704 power take-off (PTO) opening at the right or left side, or as specified (see 6.1).

3.16.2.1 Transmission, manual. Unless otherwise specified (see 6.1), a manual type transmission shall be furnished and shall have the number of speeds and ratios as recommended by the manufacturer for the required performance, but in no event shall it have less than five forward speeds and one reverse.

3.16.2.2 Transmission, automatic. When specified (see 6.1), a continuous drive automatic transmission shall be furnished of either (a) stepped ratio type with at least four forward gear speeds and one reverse; or (b) infinitely variable characteristics in forward and reverse. Means shall be provided to prohibit cranking of engine except when transmission is in neutral or park position.

(a) Stepped ratio type. The stepped ratio type transmission shall utilize a single stage, multiphase, three element hydraulic torque converter; constant mesh planetary gearing; a hydraulic control system and oil cooler; and shall be hydraulically actuated by multiple plate clutches that automatically compensates for wear. For off-highway or heavy duty service, either a four (4) speed or a five (5) speed may be specified. Either a multi-speed axle or an auxiliary transmission shall be furnished with the four or five speed automatic transmission for off-road use when specified. The five speed automatic transmission with five selector positions shall provide no less than four automatically stepped gear ratios without movement of the selector. (See 6.1).

(b) Infinitely variable type. The infinitely variable type transmission shall utilize a combination of fixed and variable hydraulic units, along with necessary internal gearing, control system, hydraulically activated wear compensated clutches, and provision for connection to a suitable oil cooling

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system. In the case of trucks designed for off-highway or heavy-duty use with infinitely variable transmission, possible need for a multi-speed axle shall be based on the specific vehicle startability requirements. Drive gears for the PTO shall be provided. When PTO is installed on the vehicle, a caution plate or decal described under 3.3.1 shall be installed in the cab on dash panel.

3.16.3 Drive line components. The drive line components shall be of sufficient capacity and design to transmit the maximum torque of the engine developed through the lowest intervening gear reduction.

3.16.4 Axles. Vehicles shall be equipped with the manufacturer's standard or optional front and rear axles. The axle capacity ratings shall be at least equal to the gross load imposed at the ground when the vehicle is evenly loaded to its rated G.V.W.R. The rear axle torque input capacity shall be at least equal to the total engine torque developed through the lowest intervening gear reductions. Gear ratios shall provide a minimum performance as specified in 3.8. Optional capacity axles may be specified (see 6.1).

3.16.4.1 Two-speed rear axles. When specified (see 6.1), or when required in combination with the transmission to provide the required performance, multi-speed axles shall be furnished and equipped with electric or air shift, operable from the driver's position.

3.16.4.2 Tandem rear axles. When a 6X4 is specified (see 1.2.1), tandem rear axles shall be furnished. Both axles shall be live and provided with power divider including inter-axle differential with lock-out control or automatic locking inter-axle differential. A warning light mounted in the cab shall be provided when a manual lock-out control is furnished. Bogie design shall be such as to permit a difference in vertical movement between any two wheels of not less than 8 inches. The torque reaction from each drive wheel shall be transmitted directly to the frame, not from one axle to the other.

3.16.5 Differential(s). When specified (see 6.1), differential(s) shall be special traction type.

3.17 Suspension. Vehicle shall be equipped with a suspension system of adequate strength to sustain the G.V.W.R. imposed by the evenly and fully loaded vehicle without evidence of overload or permanent set. If spring suspension is employed, auxiliary rear springs shall be provided (except 6 x 4's), unless progressive or variable type rear springs are furnished as standard. When specified (see 6.1), auxiliary rear springs shall be furnished regardless of type springs furnished as standard.

3.17.1 Spring stops. The manufacturer's standard spring or axle stops shall be provided. The front spring stops shall prevent the front axle from striking the engine or oil pan under all conditions of operation.

3.17.2 Shock absorbers. Heaviest-duty hydraulic, double-acting, shock absorbers available for chassis model and axle offered shall be furnished on front, except vehicles furnished with front axles rated in excess of 12,000 pounds.

3.18 Wheels, rims, tires and tubes.

3.18.1 Wheels and rims. Unless otherwise specified (see 6.1), vehicles shall be equipped with single front and dual rear wheels. Wheels and rims shall be the manufacturer's recommended size and capacity for the gross axle weight ratings. Rim contours and sizes shall conform to the current recommendations of the Tire and Rim Association, Inc.

3.18.2 Tires. Unless otherwise specified (see 6.1), all tires furnished including the spare (see 3.18.4), shall be truck type, tube or tubeless, with standard highway

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tread design on front and nondirectional nonmilitary (cross bars at right angle to center line of tread not permitted), mud and snow type on rear. Each tire shall have a load carrying capacity at least equal to the maximum load imposed upon it by the evenly and fully loaded vehicle. Tires shall be the manufacturer's recommended size and load range for the gross axle weight ratings.

3.18.3 Inner tubes. When tube type tires are furnished, the inner tubes shall conform to ZZ-I-550, except when steel-belted radial ply tires are specified. Tubes and flaps furnished shall be in accordance with recommendations of the Tire and Rim Association, Inc.

3.18.4 Spare wheel and tire. When specified (see 6.1), one inflated spare wheel or rim, tire and tube (if applicable), shall be furnished with each vehicle. Size and tread shall match that provided on rear of truck.

3.18.4.1 Spare wheel and tire mounting.

3.18.5 Tire chain clearance. There shall be no interference which will prevent use of dual triple trace tire chains on dual driving wheels in accordance with SAE J683.

3.18.6 Wheel-tire balancing. All wheel-tire assemblies of the vehicle shall be balanced in accordance with current manufacturer's standard.

3.19 Brakes. Full air brakes or vacuum booster hydraulic brakes shall be furnished as specified (see 6.1).

3.19.1 Service brakes.

3.19.1.1 Air brakes. When specified (see 6.1), full air brakes shall be furnished which conform to Federal Motor Vehicle Safety Standard No. 121 and the following requirement:

1. Automatic moisture ejectors, when specified (see 6.1).

3.19.1.2 Hydraulic vacuum booster brakes. When specified (see 6.1), the vehicle shall be equipped with vacuum or hydraulic assisted, hydraulic actuated four-wheel service brakes, with a minimum 1000 cubic inch total vacuum reserve, and low vacuum, in cab, warning indicator light or device on vacuum assisted brake system.

If a split or dual hydraulic brake system is supplied, the system shall be of such design that rupture or failure of an actuating-pressure component in the system shall not result in complete loss of function of the total system when specified. This hydraulic brake system shall be equipped with an electrically operated red light, mounted on the instrument panel to indicate system effectiveness. The indicator light shall include a means of assuring the operator light bulb is operable. "Pressure component" is any internal component of the brake master cylinder or master control unit, wheel brake cylinder, brake lines and hose, or equivalent, except vacuum assist components.

3.19.2 Brake performance. The parking brakes shall be capable of controlling and holding the vehicle in either direction, forward or reverse, loaded to its G.V.W. on a 20 percent grade, free of ice and snow. Service brakes shall conform to requirements of SAE J992.

3.20 Controls and mechanisms.

3.20.1 Controls. Unless otherwise specified (see 6.1), all controls, operating mechanisms and instruments shall be located for left-hand drive. Controls shall be complete and conveniently accessible to the operator. Lever controls shall be designed and located so as to permit easy entrance and exit to and from the operator's compart-

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ment. Instruments and controls shall be identified as to function and installed in a manner facilitating removal or servicing.

3.20.2 Steering mechanism. Unless otherwise specified (see 6.1), the chassis shall be equipped with manufacturer's standard power steering mechanism. The steering mechanism shall provide for easy operation and shall be capable of controlling the direction of the fully loaded vehicle under all operating conditions. Means shall be provided for rapid and easy adjustment of lost motion in the steering mechanism. Adequate means of lubricating the system shall be provided.

3.21 Electrical.

3.21.1 Equipment and lighting. Vehicle shall be equipped with complete 12 volt electrical generating, starting, lighting, and ignition systems, when applicable. The electrical supply source shall be as specified in 3.21.2 and 3.21.3. Lighting, signalling, wiring and protective devices, shall be provided and shall be the manufacturer's standard in conformance with Federal Motor Vehicle Safety Standard No. 108 and applicable Federal Motor Carrier Safety Regulations.

3.21.2 Batteries. Unless otherwise specified (see 6.1), the truck manufacturer shall furnish a battery or batteries which meet or exceed the engine manufacturer's recommendations for starting at temperatures of down to -10° F. In any case, following is the minimum ampere hour capacity each at a 20-hour rate.

Voltage	Gasoline	Diesel
12	120	200

When specified, a dual battery system that will provide the ampere hour capacity requested, with a multiple pole switch to provide switching arrangements conforming to Par. 1 - 2.3.1 of NFPA No. 1901

a. for ignition and lights alone
b. for starting alone
c. for starting, ignition and lighting

shall be furnished. The battery installation shall conform to Federal Motor Carrier Safety Regulation 393.30.

3.21.3 Alternator. Unless otherwise specified (see 6.1), a 105-ampere minimum alternator of 12-volt potential, equipped with voltage control, shall be provided.

3.21.4 Ignition system. Spark ignition system shall be furnished with the manufacturer's current standard ignition system with 12-volt potential. Ignition system shall be moisture proof; wiring shall be routed safely, fastened securely, clear of hazards, and shall be accessible. Ignition switch furnished shall include an accessory position. When specified, a dual ignition system shall be provided, if available (see 6.1).

3.21.5 Starter system. An ignition key switch or key switch and button starting shall be furnished with two keys. When specified, a keyless ignition switch shall be provided (see 6.1). When a non-removable key is specified, a key chained to dash is acceptable. A high torque; minimum current draw starter motor of heaviest duty available for the model and engine offered shall be furnished with compatible solenoid and wiring.

3.21.6 Wiring. Wiring shall be harness type, grease and oil resistant, moisture proof, routed in protected locations neatly and securely fastened, and all apertures properly grommetted for passing wiring. Each wiring circuit except starting and ignition circuits shall be protected by fuses, breakers, or fusible links.

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3.22 Electromagnetic interference suppression. Vehicle shall be electromagnetic interference suppressed in accordance with the manufacturer's standard practice. When specified (see 6.1), vehicle shall be suppressed in accordance with SAE J551 and J552.

3.23 Chassis equipment. The chassis equipment, etc., shall be complete and shall include, but shall not be limited to, the equipment specified herein.

3.23.1 Bumpers. Front bumper shall be provided for all vehicle types, bumper mounting brackets shall be attached to frame. Rear end protection or bumpers conforming to Federal Motor Carrier Safety Regulation 393.86, as applicable to installed body shall be furnished.

3.23.2 Fenders. Unless otherwise specified, front fenders and running boards shall be provided on all cab types. Rear fenders are not required if wheels are amply covered by equipment cabinets, and the rear wheel wells are provided with inner shields.

3.23.3 Engine hood. The engine hood on conventional cabs and cab forward type shall be so fitted to prevent water leakage from precipitation, and shall be readily removable, or shall open sufficiently to provide easy access to components in the engine compartment requiring routine maintenance.

3.23.4 Cab. Unless otherwise specified (see 6.1), manufacturer shall furnish any standard full-width conventional type cab for vehicle so equipped. Cab shall be of all metal, welded and/or riveted construction or reinforced fiberglass construction, consist of and be equipped with, but not limited to, the following: Manufacturer's standard insulation and sound-deadener; floormats; crank-operated windows; door checks and locks, operable from the inside of the cab through mechanical linkage, with at least the curb side door equipped with external key-operated lock; keys, two sets; heaviest-duty vinyl upholstering; full-width adjustable seat, or individual adjustable driver's seat and individual passenger seat; rear cab window; glove compartment; dome light or interior lighting. Cab and doors shall have adequate sealing to prevent entry of water and dust into cab. When furnishing a tilt-type cab, mechanical or hydraulic tilting and locking mechanism (in opened or closed positions) shall be furnished.

3.23.5 Seatbelts and anchors. Seatbelt assemblies, attachment hardware, and retractors for the front outboard belts shall be furnished, for driver and each occupant.

3.23.6 Glass. Glazing shall be clear and conform to Federal Motor Carrier Safety Regulation 393.60.

3.23.7 Instruments, lights, etc. The instruments shall be the manufacturer's standard easily accessible for maintenance, and shall include, but not be limited to, panel lights, indicating speedometer and recording odometer accurate within 5 percent, lighting control switches and indicators, choke when applicable, fuel shut-off for diesels, low air warning devices when air brakes are furnished, tachometer, ammeter, or voltmeter, oil pressure, engine temperature and fuel gauge(s), and heater-defroster controls. Other instruments shall be furnished, when specified.

3.23.8 Accessories. Unless otherwise specified (see 6.1), the accessories furnished shall be the manufacturer's standard and shall include, but not be limited to, dual air or electric, two or more speeds, constant in each speed, windshield wipers; horn(s); heater and defrosters; dual sun visors; ashtray, cigar lighter; dual outside rearview mirrors; back-up warning device.

3.23.8.1 Mud flaps. Mud flaps shall be furnished on all vehicles when the horizontal distance between the center line of rear wheels and the rear of vehicles is less than three times the vertical height from ground to end of rear platform or body.

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Mud flaps furnished and installed shall be full-width of dual rear tires, conforming to SAE J682 and devoid of advertisement, except for supplier's name or emblem, twist and sailproof.

3.23.8.2 Mirrors, rearview. Outside rearview mirrors furnished shall be dual western type swing or extendable, bracket mounted, independently adjustable mirror heads of the largest size appropriate for vehicle model offered. Mirror heads shall incorporate a separately adjustable wide angle (convex) section. Mirrors shall be securely mounted to cab or cab doors and free of vibration. Reflective area shall be not less than 70 square inches each.

3.23.8.3 Heater and defroster. Hot water, fresh air type heater and defroster with a 20,000 B.t.u. per hour minimum rating sufficient for occupant's comfort with dash mounted panel controls convenient to the driver, shall be provided. Discharge outlets shall be provided to direct heater air to floor and to defroster louvers by blowers. When right-hand drive vehicle is specified, recirculating type heater and defroster may be furnished in lieu of fresh air type. Cab compartment ventilator system other than windows shall be provided on all vehicle models. A larger capacity heater or an auxiliary recirculating heater may be specified for 5 man enclosed or canopy cabs.

3.24 Tools. When specified (see 6.1), each vehicle shall be furnished with the manufacturer's standard tools including, but not limited to, hydraulic jack of sufficient capacity to lift wheel of fully loaded vehicle to height necessary to permit removal and replacement of wheel and tire assembly, jack handle, and lug nut wrench.

3.25 Body and apparatus.

3.25.1 Body. Unless otherwise specified (see 6.1), the truck shall be equipped with a body which shall incorporate a pump housing, operator's panel, hose bed, equipment cabinets, running boards, rear step, and water tank. Four folding steps shall be furnished, two at rear and two at front, on right and left side of vehicle, conveniently placed to provide access to hose bed or reel(s). The body shall be adequate for mounting the fire pump and stowing the fire fighting equipment specified herein. The body shall be made of not less than 0.1046-inch thick steel and shall be reinforced at all points where hose and ladder clamps and hangers are attached.

3.25.2 Pump housing and stowage bin. Unless otherwise specified (see 6.1), a top stowage bin shall be provided over the pump. This bin shall have a removable steel panel bottom large enough to enable the pump to be removed with the suction and discharge connections removed. The stowage bin bottom shall support 200 pounds on any one square foot area without permanent deformation. The sides of the bin shall be the same height as the sides of the hose bed. Adequate drain holes shall be provided to preclude water retention.

3.25.3 Fire hose compartments. Unless otherwise specified (see 6.1), a hose compartment(s) to be provided, designed to accommodate not less than 1,500 feet of 2-1/2 inch rubber lined double jacketed fire hose, and 400 feet of 1-1/2 inch rubber lined, double jacketed fire hose. A partition shall be provided to separate the 1-1/2 inch and 2-1/2 inch hose. Compartments shall be constructed of steel and reinforced at corners. Flooring to be constructed of well seasoned, select grade of oak or other hardwood, treated to prevent damage to hoses. Slats to be no less than 3-1/2 inches wide x 3/4 inch thickness, spaced to provide ventilation. Flooring shall be removable.

3.25.4 Equipment cabinets. Unless otherwise specified (see 6.1), two equipment cabinets shall be provided on each side of the body, front and rear of the wheelhousing, approximately flush with the running boards and the wheel skirting. The cabinet width shall be approximately 20 inches; height from running board to the top of the wheelhousing shall not be less than 25 inches and depth shall be the same as the wheelhousing but not less than 20 inches. Each cabinet shall have a piano hinged forward-type door, gasketed to prevent the entry of water into the compartments, and a locking

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mechanism. Each cabinet door shall be provided with a recessed paddle handle lock, or recessed "D" ring type, with latch at top and bottom of door. The clear opening for these doors shall be not less than 18 inches wide and 23 inches high. Cabinets shall be provided with shelves when specified. In addition, a cabinet shall be provided at the rear of the truck. The width of the cabinet shall be such that the cabinet will fit between the truck frame; the height shall be such that the cabinet will fit between the rear step and the bottom of the hose bed; and the depth shall be approximately 12 inches. The door(s) shall be hinged at the side or bottom and shall be equipped with latches identical in exterior appearance to those installed on the side cabinets. All cabinets shall be illuminated. Cabinet lights shall automatically be controlled through door switches to light when cabinet doors are opened. Cabinet tops and wheel-housing tops shall run continuously on each side of the truck and shall be of antiskid surface similar to running board and rear step surfaces. Drip molding shall be provided over cabinet doors.

3.25.5 Running boards and rear platform. Unless otherwise specified (see 6.1), the truck shall be equipped with running boards and a rear platform. The rear platform shall be of a size to accommodate four men, weighing 175 pounds each, without noticeable deflection of the rear step. The running boards and platform shall be constructed of sheet steel of not less than 0.1196 inch thickness and shall have a nonslip tread surface. The running boards and rear platform shall be not more than 24 inches above ground level. The rear step shall have a radius at the corners and shall be flanged downward at least 2 inches. Side running boards shall be continuous between the rear of front fenders and front bulkheads of forward equipment cabinets, and shall be of the same material and equivalent strength as the rear step.

3.25.6 Handrails. Unless otherwise specified (see 6.1), the rear step shall be provided with two vertical grab rails for mounting the step, and a horizontal handrail not over 66 inches above the step. The handrail shall be located not more than 6 inches forward of the rear end of the hose bed, or 6 inches forward of the clear area of the rear step. Hose pickup lights shall be mounted at each end of the handrail. Both the grab rails and handrail shall be not less than 1-1/4 inch heavy chromium plated brass, or stainless steel tubing and shall be securely mounted. Any equipment attached to the body within reach of a running board shall be securely attached so as to serve as a personnel handrail.

3.25.7 Booster tank. The tank shall be of rectangular design with a removable top. The top to be large enough to permit complete cleaning or repair of tank or baffles. The top shall be gasketed to prevent leakage due to surging of water, the top retaining screws or fasteners shall not come in contact with the water. Tank construction shall allow water to be pumped out when the truck is standing on a 20% grade in all planes. Internal baffles to be provided. The tank shall be equipped with suitable method for lifting tank out of body. The water tank shall be constructed of stainless steel if specified (see 6.1), or of high strength, corrosion resistant, low alloy, copper bearing steel, minimum copper content 0.25-0.55 percent - minimum chromium content 0.30-1.25 percent, the interior and exterior of the tank shall be suitably protected against rust, corrosion and deterioration. The material used shall be a minimum of 0.1345 inches with a minimum yield strength of 50,000 psi. The tank shall be provided with a convenient fill opening of not less than 5 inches in diameter and properly protected by a strainer which is secured with provisions for removal to facilitate cleaning. A closure to prevent water from splashing out while vehicle is in motion shall be furnished, with quick release for opening. Three inch air vent and overflow provided to protect tank against damage while pumping from tank with filler cap in closed position. Piping between the tank and pump suction and all necessary valves, fittings, drains, etc., provided and be vibration proof construction. Flexible connections provided in discharge and suction lines between tank and chassis or body. All operation valves shall be ball type, quick opening, with controls mounted on operator's panel on left side of truck. Tank capacity shall be specified (see 6.1). Provide drains in all piping where water may freeze if trapped, and accessible from the operator's side of truck without the necessity of crawling underneath the truck, drain control rod shall

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not hang below frame of truck. Outlet connections between tank and pump shall permit a flow of not less than 250 gallons per minute.

3.25.8 Piping. Booster suction piping from tank to pump to be 2-1/2 inches I.D. pipe minimum with flexible fittings and full flow ball quarter turn booster tank valve with control mounted on operator's panel. Tank connection to have 8 inch swirl baffled plate above tank outlet. Piping provided with proper drain. A 1-1/2 inch inside diameter tank refill line with control valve shall be provided for filling the tank from the fire pump. The valve control handle shall be extended to the pump operator's position. All piping, unless made of corrosion-resistant metal, shall be galvanized. The pipe line between pump and booster reel(s) shall not be less than 1 inch inside diameter and shall incorporate a 1-1/2 inch quick opening shut off valve with control extended to the operator's control panel. Couplings, screw threads, and gaskets shall be in accordance with National Fire Protection Standard No. 194. All ball valves to be designed to be taken apart for service without disturbing piping, if other than specified (see 6.1).

3.25.9 Fire pump. The pump shall be mid-ship mounted, multi-stage centrifugal parallel-series type, or a single stage type when specified, having a rated capacity as specified (see 6.1), driven by the truck engine from the output shaft of the road transmission. The pump casing shall be ferrous alloy with a stainless steel shaft and bronze impeller(s). Pump to be ball or roller bearing type, use of an internal sleeve type impeller shaft bearing is acceptable; with impeller(s) to be on both static and running balance. The pump transmission case shall be two piece type of ferrous alloy; the gears shall be of hardened alloy steel; the pump transmission shall be equipped with an enclosed circulating water-oil cooler if it is needed to maintain the oil operating temperature recommended by the pump manufacturer. All levers or other devices used in connection with pumping operation shall be provided with positive catches or other means to prevent them from being jarred or moved out of position. All controls for changing into pumping mode shall be operable from driver's position and shall be adequately marked for pump and road position, if other than specified (see 6.1).

3.25.10 Priming pump. The priming pump shall be of the rotary type, 12-volt, motor-driven, complete with lubricating oil reservoir and necessary valves, operated from the pump panel. The priming pump shall develop a vacuum of 22 inches of mercury (Hg) at an altitude up to 1,000 feet above sea level in not more than 30 seconds and shall hold the vacuum with a drop not exceeding 10 inches of Hg in 10 minutes with the pump shut off, if other than specified (see 6.1).

3.25.11 Pressure relief valve. An automatic relief valve or pressure regulator shall be provided for controlling the pressure from all outlets. The location of the relief valve or pressure regulator control shall be such that adjustment can readily be made by the operator when standing in normal position, without the use of tools, and in full view of all discharge pressure gages. The relief valve or regulator shall be of such design that when set 10 psi higher than the pump working pump working pressure, there will not be an increase in pump pressure greater than 30 psi when discharge lines are shut slowly. The relief valve or regulator shall perform satisfactorily when pumping rated capacity at any pump working pressure over 75 psi, if other than specified (see 6.1).

3.25.12 Transfer valve. The transfer valve shall be electrically or hydraulically operated. The position for parallel (volume) and series operation (pressure) shall be clearly indicated along with a manual override provided on operator's panel. The transfer valve shall have the ability to change from series operation to parallel without reducing engine speed maintaining an effective fire stream at the nozzle at all times, if other than specified (see 6.1). A manually operated transfer valve may be furnished, when specified.

3.25.13 Auxiliary cooling system. A supplementary heat exchange cooling system shall be installed to permit use of water from the discharge side of the fire pump for

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cooling of water circulating through the engine cooling system. Heat exchanger to be of bronze and copper construction and shall be a separate unit installed in pump compartment with valve control on operator's panel. An emergency radiator refill line shall also be provided with control on operator's panel, when specified (see 6.1).

3.25.14 Suction inlets. Suction inlets shall be provided as specified (see 6.1), each inlet shall be provided with a long handled chrome plated cap. Bleeder drain valves shall be provided on gated inlets. All inlets shall have removable strainers.

3.25.15 Discharge outlets. Discharge outlets shall be provided as specified (see 6.1), each outlet shall be provided with a drain valve for draining hose lines with pump discharge gates closed. Each discharge outlet shall be provided with a rocker lug, chrome plated, gasketed cap chained to the valve body. There shall be provided a 1-1/2 inch pipe for through connections from pump to rear of hose bed, mounted in rear body panel to supply 1-1/2 inch preconnected hose line, unless otherwise specified. Preconnect line shall be controlled by a quarter turn ball type valve, located on operator's panel.

3.25.16 Pump equipment. Unless otherwise specified (see 6.1), the following equipment shall be furnished.

(a) Two ten foot lengths of soft suction hose to be furnished, size as specified (see 6.1), complete with female NST, unless otherwise specified (see 6.1), couplings and mounting brackets.

(b) Two ten foot lengths of hard suction hose, size as specified (see 6.1), complete male and female couplings mounted in steel troughs or tubes, threads to be NST, unless otherwise specified, with chrome plated quick release locks.

(c) One aluminum suction hose strainer, size as specified (see 6.1), and thirty feet of rope with mounting bracket.

3.25.17 Hose reel(s). Unless otherwise specified (see 6.1), high pressure, drum type, hose reel(s), electric, 12 volt, rewind type gear or chain driven drum, complete with 200 feet of one inch, 800 pound working pressure hose (four 50-foot lengths with "Barway" couplings and high pressure fog nozzle(s) shall be furnished). The hose reel(s) shall be mounted above the pump at the forward end of the body. Chrome plated guides or rollers shall be provided, one on each side of reel opening. Reel(s) shall be equipped with positive locking brakes so as to prevent loosening on the reel drum during a pumping operation. A hand rewind crank shall be provided for retrieving of hose in the event of an electric failure. Valve(s), (quarter turn ball type), serving hose reel(s) shall be controlled from pump panel. The hose reel swivels furnished shall be of the non-packed type water lubricated. The reel and swivels shall be capable of withstanding 1,000 psi static pressure test. The fog nozzle(s) shall be capable of delivering no less than 30 gallons of water per minute and have a complete range of adjustment from full fog to long range stream. The adjustment shall remain constant after being set, not dependent on grip of hands to keep the shape of fog required. Shut-off control shall be separate from fog adjustment. Two Bar-way coupling spanners shall be furnished. The reels provided shall be of current commercial manufacture.

3.25.18 Operator's control panel. If other than specified (see 6.1), all pump controls and gauges to be located at left side of apparatus and properly marked. Pump panel shall be satin finished stainless steel. Embossed metal plates, permanently secured with fasteners, shall be provided indicating each control on panel. Discharge controls shall be either push-pull or swing type. Push-pull discharge controls to have chrome or cadmium plated rods, minimum 1/2 inch diameter. Discharge controls to be locking type, push-pull type to pull straight out of panel. Remote control push-pull rods shall be equipped with guides and joints designed to eliminate binding. Pump panel shall be designed to permit ease of removal. Ample lighting shall be provided so that the operator's panel will be fully illuminated with a full length polished stainless steel shield provided to direct the light downward and protect the bulb(s).

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The following controls and gauges shall be provided, mounted on the panel:

(a) Suction inlet controls.
(b) Discharge outlet controls, 1/4 turn ball type valves.
(c) Relief valve control.
(d) Controls for tank refill or high pressure bypass.
(e) Manual pump shift overide control.
(f) Master pressure and suction gauges, 30 in. Hg.-0-600 lb./sq. in.
(g) Individual pressure gauges for all 2-1/2 inch discharges 30"-0-600 lbs., 3-1/2 inch diameter.
(h) All gauges shall have snubbing devices or pinch cocks and test plugs.
(i) Engine oil pressure gauge.
(j) Engine water temperature gauge.
(k) Tachometer, a handcounter r.p.m. outlet and engine hour meter.
(l) Master pump drain control.
(m) Individual drain valves.
(n) Tank water level gauge.
(o) Vernier type quick action engine throttle.
(p) Manufacturer's and underwriter's data plate.

3.25.19 Electrical equipment.

(a) Two chrome plated, sealed beam, inner cab controlled spotlights, 6 inch size, mounted on cab.
(b) Two swivel-type, 6 inch seal beam, chrome plated, hose pickup lights mounted, one on each rear corner, above the hose compartment.
(c) Two back-up lights, flush mounted in rear of body, to provide ample illumination for backing up vehicle.
(d) Engine compartment work light.
(e) Pump compartment work light.
(f) Combination tail, stop, and directional signals.
(g) Clearance and marker lights, reflectors, and license plate bracket and lights shall conform to FMVSS No. 108. The rear cluster marker lights shall be mounted in a manner and location to provide for protection of lights.
(h) Two red warning lights shall be provided at rear, mounted on brackets, wired to flash alternately, wiring to be completely enclosed, this bracket to also support the two hose deck lights.
(i) A master switch shall be provided.
(j) A polarized battery charging receptacle shall be provided.
(k) A 12 volt emergency revolving light shall be furnished, having 4 seal beams, 2 clear and 2 red, each clear lamp having a minimum of 35,000 candle power, mounted on top of cab and shall be visible at all times through 360 degrees. The control for the warning light shall be independent of the siren and shall be provided with a pilot light indicating the "on" position. The pilot light shall be located in the cab on the instrument panel.
(l) When specified (see 6.1), a 110 volt AC or DC on board power system shall be provided for the operation of tools and rescue equipment, with required outlets for plug-in of this equipment.

3.25.20 Communications.

(a) Radio (mobile) provisions. All fire engines shall be provided with sufficient ventilated space for a modern two-way radio, convenience features (antenna openings, ground plate, terminal wiring and panels, etc.), required for the installation, and servicing of mobile radios. Unless otherwise specified (see 6.2), the consignee will locally install mobile radio unit(s) or shall provide any additional data, equipment, and a coordinator to insure compatibility of specified systems.

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(b) Antenna, cable and mountings. Antenna when furnished shall meet the requirements of EIA Standard RS329 Part II. The coaxial lead-in wire furnished shall be of the impedance, loss, and power handling capability specified by the mobile radio manufacturers. The contractor shall provide each fire engine a ground plate and opening, and coaxial lead-in wire from the ventilated radio storage compartment to the antenna port.

(c) Two-way (mobile) radio equipment. When specified (see 6.2), a solid state (no tubes) mobile two-way radio shall be furnished, with all normal accessories. This radio shall meet the requirements of EIA Standards RS152B (transmitters), RS204A (receivers), RS329 Part II (mobile antennas), and, if applicable RS220 (continuous tone coded squelch systems) and RS374 (selective signalling). The transmitter power output shall be *(as specified) watts minimum. A noise cancelling microphone shall be located in the driver's compartment. Receiver audio power output shall be *(as specified) watts minimum. Selectable speaker outputs, singularly and in combination shall be provided at the driver's position, and through the public address system. Radio interference suppression (see 3.22). *Power to be specified (see 6.1), consistent with local emergency communication requirements. All switches and control devices shall be clearly visible for personnel operating the equipment; identified plainly and permanently marked; grouped according to function, mounted in lighted panels, or consoles and installed in a manner to facilitate removal and servicing. All controls shall comply with FMVSS 101a.

(d) A signalling device, a buzzer or similar device for communicating between cab and rear step shall be furnished.

3.25.21 Siren - public address system. The electronic siren and public address system furnished shall include radio amplification capability, complete with speaker(s) and microphone. The siren control functions and related characteristic sound patterns shall consist of manual and automatic wail, yelp, and "Hi-Lo" sound when applicable or specified. Plug-in connections and connecting cables, foot switch and all hardware necessary for normal siren/speaker(s) operation shall be furnished. The siren unit shall have an amplifier capable of producing continuous warning tones with a power output of not less than 80 watts RMS with one speaker load and not less than 130 watts RMS with a two speaker load when the amplifier is supplied with 13.6 volts dc input. The speaker(s) furnished shall have a minimum 100 watt RMS rating. In voice operation, the power output shall be a nominal 35 watts, with not more than 10 percent voice distortion from 300 to 3000 Hz. The public address amplifier shall be independent of the two-way (mobile) radio except that a common microphone and control housing group may be employed. The siren assembly (including parts installed in the passenger space of the emergency vehicle) shall pass the radio noise interference tests specified in SAE Standard J551B. One or two (siren/public address/radio) speakers as required shall be installed outside on the vehicle. Wherever practical, metals used in manufacturing shall be either corrosion-resistant (polished metal finish) or treated to resist corrosion (painted white). The use of dissimilar metals, especially brass, copper or steel in intimate contact with aluminum or aluminum alloy shall be avoided wherever practical. Design and construction shall ensure that no parts will become loose and shall withstand the strains, jars, vibrations, shocks and other similar conditions incident to service operations. Installation shall be on reinforced mounting to prevent drumming and vibration. The siren/speaker manufacturer shall provide detailed instructions for installation. The speaker(s) shall be mounted facing forward, located near the longitudinal centerline and front section of the emergency vehicle. Speaker location shall permit servicing of the vehicle without removing the speakers or interfering with the intent or vision of the warning light(s). If dual speakers are employed, the manufacturer shall provide the following information:

a. The angle between the acoustic axis of each speaker and the longitudinal center-line of the vehicle.

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b. Electrical connections specifying whether the speakers are connected in phase or out of phase.

c. Directionality plots or directionality response data for each suggested mounting configuration, when applicable.

Electronic sirens, amplifier boxes, drivers and speakers shall be permanently marked with the manufacturer's name, initial or lettered trademark and the model designation in letters and numerals nominally 1/8-inch height. The power (wattage) rating of the speaker driver specified by manufacturer of the driver shall be clearly marked on the driver. The normal operation of the siren shall be such that the power delivered to the driver shall not exceed the power rating of the driver. An electric-mechanical 12 volt siren, heavy duty, coaster type, chrome plated, electric brake, capable of producing 120 dba shall be provided when specified. Siren to be operated by foot switches, one mounted on each side of cab.

3.25.22 Accessories. Unless otherwise specified (see 6.1), the following shall be furnished and mounted on fire truck:

a. One chrome plated 2-1/2 inch ball type shut-off nozzle and play pipe assembly; nozzle to have 1 inch and 1-1/8 inch tips.

b. Removable nylon or canvas cover(s) for hose and stowage compartment(s).

c. One ten foot pike pole with wood or fiberglass handle.

d. One 15 lb. CO_2 fire extinguisher.

e. One 20 lb. dry chemical extinguisher.

f. One 2-1/2 inch double female connection, chrome plated.

g. One 2-1/2 inch double male connection, chrome plated.

h. Two metal screw flanges for play pipes.

i. Two electric hand lanterns, fire department type, utilizing 2-6 volt lantern type dry cells, minimum 4-1/2 inch diameter reflector, a main bulb shall produce a long range beam, a smaller auxiliary bulb consuming less power shall be provided for prolonged or emergency use with a switch controlling either long range or emergency mode, provisions for switching from operating battery to reserve. Battery is required, the battery case and lamp shall be water and rust proof and mounted with bracket.

j. One six pound flathead axe, secured in chrome plated mounting.

k. One six pound pickhead axe, secured in chrome plated mounting.

l. One 36 inch steel crowbar, secured in chrome plated mounting.

m. One 42 inch Hooligan tool, secured in chrome plated mounting.

n. Unless otherwise specified (see 6.1), a 35 foot three section aluminum fire ladder, solid side, to be provided and mounted. Ladder to have full length engagement between sections. Engaging surfaces to be leveled to provide self-centering action when ladder is used. Rungs to be expanded into rectangular box section side rails by means of an expander sleeve or bushing.

o. Unless otherwise specified (see 6.1), a 14 foot aluminum roof ladder, fire, with folding hooks, shall be furnished with rungs expanded into rectangular, box section side rails by means of an expander sleeve or bushing.

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p. Unless otherwise specified (see 6.1), a 10 foot folding aluminum attic ladder shall be furnished.

q. Unless otherwise specified (see 6.1), 1,500 feet of double jacketed cotton-polyester 2-1/2 inch fire hose of 50 foot lengths, coupled by light weight forged couplings in accordance with Fed. Spec. WW-C-621E, installed in hose bed, shall be furnished.

r. Unless otherwise specified (see 6.1), 400 feet of double jacketed cotton-polyester 1-1/2 inch fire hose of 50 foot lengths, coupled by light weight forged couplings in accordance with Fed. Spec. WW-C-621E, installed in hose bed, shall be furnished.

s. Two 1-1/2 inch shutoff nozzles capable of discharging a minimum of 100 gpm in both a spray and straight stream.

t. Two 2-1/2 inch shutoff nozzles capable of a minimum of 250 gallons in both a spray and straight stream.

3.26 Name and caution plates. Manufacturer's standard production data, caution and instruction plates for both chassis and fire apparatus. For chassis detail, see paragraph 3.5.2.

3.27 Preparation for painting. All chassis and body surfaces concealed or exposed, except polished metal parts that are corrosion resistant or plated, shall be cleaned, treated, and coated with a firm primer or other preservatives with rust-inhibiting properties for quality protection. Subsequent preservatives or finishes shall be industry's standard practice.

3.27.1 Paint finish and color. Unless otherwise specified (see 6.1), the exterior color shall match the color chip of paint number 11105 red, of Federal Standard 595a. The interior finish shall be the same color as the exterior or a compatible contrasting color.

3.28 Special requirements. Specify (see 6.1), unusual operating conditions, items, exceptions, end use, etc., not specified herein. If in conflict with the manufacturer's standard equipment and options, those specified shall take precedence.

3.29 Workmanship. Vehicles shall be free from defects which may impair their serviceability or detract from their appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 A representative of Underwriters Laboratories, Inc. shall conduct an inspection when requested by GSA - Quality Assurance Specialist, or when UL certification is required for assurance of community insurance rates; otherwise, the contractor may test and shall issue certification or proof that the UL factory test requirements for fire department pumpers has been met. This document shall be furnished to the ordering agency at time of delivery by the contractor, along with "Manufacturer's Record of Pumper Details" and other certification and details required by NFPA, Standard No. 1901.

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5. PREPARATION FOR DELIVERY

5.1 Preparation. Unless otherwise specified (see 6.1), the truck(s) shall be preserved and packaged for mobile delivery in accordance with the supplier's standard commercial practice, insuring carrier acceptance and safe delivery to destination in compliance with regulations applicable to the mode of transportation.

6. NOTES

6.1 Ordering data. Purchaser should select the preferred options permitted herein and include the following information in procurement document:

a. Title, number, and date of this specification.

b. Type and size of vehicle, pumping capacity and tank size required to meet fire apparatus requirements (see 1.2.1).

c. Special equipment required other than specified (see 3.3.1).

d. Gross vehicle weight rating (G.V.W.R.) for fire truck(s) (see 3.5.1.1).

e. Dimensions of vehicle, CA, or wheelbase, etc. (see 3.5.3 and 3.5.3.1).

f. Frame and frame reinforcement (see 3.6 and 3.6.1).

g. Power unit, state type and/or special power or engine configuration description (see 3.7.2 and 3.7.3).

h. Oil filter, specify if different (see 3.10).

i. Air filter, if other than specified (see 3.11).

j. Emission controls, specify if California requirements are required (see 3.12).

k. Fuel tank, specify capacity and style required (see 3.13).

l. Shutters, if required (see 3.14.2).

m. Exhaust system, if location is specified (see 3.15).

n. Spark arrester exhaust, if required (see 3.15).

o. Power take-off opening if other than SAE P.T.O. opening is required (see 3.16.2).

p. Transmission, if other than specified (see 3.16.2.1).

q. Automatic transmission, specify number of speeds, if required (see 3.16.2.2).

r. Multi-speed rear axle or auxiliary transmission when required with an automatic transmission (see 3.16.2.2).

s. Optional axle capacities, when required (see 3.16.4).

t. Multispeed rear axle, when required (see 3.16.4.1).

u. Differential, special traction type, if required (see 3.16.5).

v. Auxiliary rear springs, when specified (see 3.17).

w. Wheels and rims, if other than specified (see 3.18.1).

x. Tires, if other than specified (see 3.18.2).

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y. Spare wheel and tire, specify if required and if special mounting is required (see 3.18.4 and 3.18.4.1).

z. Brakes, air or vacuum when specified or if other than required (see 3.19.1).

aa. Controls, if other than specified (see 3.20.1).

bb. Steering, if other than specified (see 3.20.2).

cc. Battery, if other than specified (see 3.21.2).

dd. Alternator, if other than specified (see 3.21.3).

ee. Electromagnetic interference suppression, if other than manufacturer's standard practice is desired (see 3.22).

ff. Cab, if other than specified (see 3.23.4).

gg. Accessories, as specified (see 3.23.8).

hh. Heater, if other than specified (see 3.23.8.4).

ii. Tools, when specified (see 3.24).

jj. Pump housing and storage bin, if other than specified (see 3.25.2).

kk. Fire hose compartment, if other than specified (see 3.25.3).

ll. Equipment cabinets, if other than specified (see 3.25.4).

mm. Running boards and rear platform, if other than specified (see 3.25.5).

nn. Handrails, if other than specified (see 3.25.6).

oo. Booster tank, specify capacity and if other than required (see 3.25.7).

pp. Piping, if other than specified (see 3.25.8).

qq. Fire pump, specify rated capacity and if other than required (see 3.25.9).

rr. Priming pump, if other than specified (see 3.25.10).

ss. Pressure relief valve, if other than specified (see 3.25.11).

tt. Transfer valve, if other than specified (see 3.25.12).

uu. Auxiliary cooling system, if other than specified (see 3.25.13).

vv. Suction inlets, specify number and size (see 3.25.14).

ww. Discharge outlets, specify number and size (see 3.25.15).

xx. Pump equipment, specify size and threads if other than specified (see 3.25.16).

yy. Hose reels, if other than specified (see 3.25.17).

zz. Operator's control panel, if other than specified (see 3.25.18).

aaa. Electrical equipment, when specified or if other than specified (see 3.25.19).

bbb. Communications (see 3.25.20).

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ccc. Accessories, specify if different (see 3.25.22).

ddd. Paint, if other than specified (see 3.27.1).

eee. Special requirements (see 3.28).

fff. Inspection, if different than specified (see 4.1).

ggg. Preparation and delivery, specify if different (see 5.1).

hhh. Illustrations and drawings, if other than specified (see 6.3.3).

iii. Truck Ability Prediction Procedure, SAE Form J688 may be omitted when so indicated in contract (see 6.2).

jjj. Manuals, if other than specified (see 6.3.5).

kkk. Driveaway delivery, if required (see 6.3.8).

6.2 Performance prediction. Complete copies of SAE J688 Truck Ability Prediction Procedure form and computation for high and low speed requirement for each vehicle model furnished under contract shall be submitted with the required drawings, unless otherwise specified (see 6.1). The SAE Work Sheet Item 1 should include vehicle model number, wheelbase, engine model number, and vehicle type. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry air. The factors to be used in predicting truck ability (see 3.8.3) are established as follows for the corresponding SAE Truck Ability Prediction Procedure Tables:

Table 1. Tire factor relates to the size of tires furnished by the contractor in accordance with this specification.

Table 2. Altitude factor - 1.00.

Table 3. Rolling factor - 1.613.

Table 4. Area factor - Use actual frontal area of fire engine presented by the chassis and with body mounted.

Table 5. Velocity factor - 250.0.

Table 6. Altitude factor - 1.00.

Table 7. Chassis friction horsepower - Use applicable power unit G.V.W. (to nearest, higher 1,000 pounds) and the engine r.p.m. (to nearest 100 revolutions) which is required for 50 m.p.h. geared speed. For G.V.W. and engine speed beyond this range of this table, factors shall be extrapolated.

Table 8. Grade factor - 0.75.

Table 8A. Correction factor - Not required.

Table 9. Road factor - 0.0.

6.3 Procurement requirements. Invitation for bids, contract, or orders should contain the following requirements (see 6.3.1 through 6.5), except in those instances where it is determined that inclusion thereof would not be in the best interest of the Government.

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6.3.1 Domestic warranty. The contractor, on vehicles used within the fifty States of the United States and the District of Columbia, hereby warrants vehicle to be free from defective material and workmanship as follows:

The entire vehicle (except tires and tubes, if warranted by their manufacturers and normal maintenance services) shall be warranted for one year from date of acceptance or 12,000 miles road travel, whichever may occur first. Engine and power train components (as covered by the vehicle manufacturer's standard warranty to the general public for the current year of manufacture), shall be warranted at 50 percent of the normal charge for parts and labor imposed by the dealer or other authorized facility from 12,001 to 50,000 miles provided this occurs within the original 12 months. Provided, however, that if the contractor receives from any supplier or subcontractor any additional warranty on the whole or any component of the vehicle, either in the form of greater time and/or mileage or otherwise and including any pro-rata type arrangement, or if the contractor generally extends to his commercial customers a greater warranty coverage than that provided the Government under this contract, the Government shall receive corresponding warranty benefits.

The warranty shall include the furnishing, without cost to the Government, F.O.B. contractor's nearest dealer or branch, or to the original destination if desired by the Government, of new parts and assemblies to replace any that prove to be defective within the warranty period. In addition, when the Government elects to have the work performed by the contractor, the cost of the labor involved in the replacement of the defective parts or assemblies at the contractor's plant, branch, or dealer facility shall be borne by the contractor. On vehicles procured F.O.B. destination and delivered by the driveaway delivery method, when authorized by the Government, warranty limitation will be in addition to the mileage accumulated by such driveaway method.

6.3.2 Export warranty. The contractor, on vehicles used outside the fifty States of the United States and the District of Columbia, warrants the entire vehicle (except tires and normal maintenance services) to be free from defective material and workmanship, for one year from date of acceptance or 12,000 miles road travel, whichever may occur first. On vehicles delivered to the Government by driveaway method, the 12,000 miles warranty limitation will be in addition to the mileage accumulated by such driveaway method. This warranty shall only include the furnishing of new parts or assemblies to replace any returned to the contractor by Government which prove to be defective. The replacement parts or assemblies shall be delivered by the contractor to the port of embarkation in the United States designated by the Government. The contractor shall not be required to bear the cost of the labor involved in correcting defects in vehicles used outside the fifty States of the United States and the District of Columbia.

6.3.3 Illustrations and drawings. Four complete sets of working drawings of the proposed vehicle(s) shall be supplied within 45 days after receipt of notice by the successful bidder, for the approval of the ordering agency. Drawings shall include plans, elevations, sections and detail drawings. All material, equipment and apparatus shown in the drawings shall be considered part of the offer and shall become part of the contract.

6.3.4 Repair parts and service. As a continuous operation of the vehicles contemplated by this specification is of utmost importance, it is necessary that the successful bidder be in a position to render prompt service and to furnish replacement parts. Accordingly bidders shall indicate the extent of their ability to render prompt service by furnishing, upon request, a list of branch offices or agencies where complete stocks of repair parts are maintained and can be secured within a reasonable time after ordering by part number from the manufacturer's parts book and at such discount as may be quoted from year to year by the manufacturer of the vehicle purchased under this specification.

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6.3.5 Manuals, operating and servicing. The successful bidder shall furnish two operator's handbooks and two lubrication charts, if the charts are not in the handbooks, with each vehicle. When specified, two spare parts lists or books and shop repair manuals for vehicle and furnished equipment, shall be furnished with each vehicle (see 6.1).

6.3.6 Loading. When shipment is to be made by rail, water, or truck, the vehicle shall be prepared and loaded for shipment in accordance with the best commercial practice. When shipment is to be made by rail on Government bill of lading, freight cars shall be loaded so that the Government will obtain the most advantageous freight rate.

6.3.7 Servicing prior to loading. Servicing prior to loading shall be in accordance with GSA Form 1455, Motor Vehicle Preshipment Inspection and Servicing Requirements. Unless shipping regulations prohibit, the crankcase shall be filled to the operating level with high quality lubricating oil of the correct viscosity for the "run-in" period as recommended by the engine manufacturer. In addition, the fuel tank shall be filled with 5 gallons of fuel, and the cooling system shall be filled with a mixture of water and ethylene glycol antifreeze (thatwill not affect aluminum blocks) in proper proportions to prevent freezing at the lowest ambient temperature enroute or at the destination. Vehicles which are shipped with drained crankcase, half-filled transmission, and dry oil bath air cleaner, shall have a warning clearly indicated on tag, securely attached to the steering wheel or clearly marked on the windshield that the crankcase, transmission, and air cleaner must be filled prior to running the engine.

6.3.8 Servicing for driveaway delivery. When driveaway by Government personnel is specified in the invitation for bid (see 6.1), the contractor shall perform all the servicing operations specified in 6.3.7.

6.4 Intended use. Purchasers should outline clearly the conditions under which the vehicles required will operate, describing the performance and ability necessary. Any unusual operating conditions that would justify deviations from standard construction as required by this specification should be specifically described in 3.28.

6.5 Statement of Origin or Bill of Sale. A manufacturer's Statement of Origin or Bill of Sale showing the applicable Delivery Order Number is required for each vehicle procured under this specification. Unless otherwise specified, such documents shall be forwarded to the consignee.

6.6 Instructions. The contractor shall provide a qualified and responsible representative to be present at time of delivery of the fire truck. The representative or delivery engineer shall remain for a period of two days for the purpose of instructing fire department and maintenance personnel in the operation and maintenance of the apparatus, unless otherwise specified (see 6.1).

6.7 Preproduction conference. When requested by the contracting officer, a meeting of the contractor, contracting officer, a representative of the ordering agency and the specification manager shall be held prior to the start of construction of the fire engine(s).

CUSTODIANS:
GSA-FSS

Preparing activity
GSA-FSS-FYS

Coordinating activities:

Military

Civil agencies:

DOT - Coast Guard
Interior - BIA - NPS
U.S. ERDA - ERDA
VA - VA (134)

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail. to: GSA-FSS-FYS-CMBg #4, Rm. 903, Washington, DC 20406

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION KKK-F-001964

Fire Truck (Pumper) Gasoline or Diesel Engine

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

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